

Cyclistic Bike-Share Analysis

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Case Study Summary

This report presents an analysis of Cyclistic's bike-share data from 2019 to early 2020. The goal was to identify behavioral differences between **casual riders** and **annual members** to support the company's marketing strategy for increasing annual memberships.

The datasets were obtained from the official Divvy Trip Data portal, containing over 4.2 million rides across five quarters. The data was cleaned, standardized, and merged into a unified dataset for consistent analysis. Error entries, missing values, and unrealistic trip durations were removed to ensure accuracy.

Analysis Highlights

User Composition: Members accounted for roughly 76% of total rides, while casual riders made up 23%.

Ride Behavior:

- Members typically rode during weekday, commute hours (7–9 AM and 5–6 PM).
- Casual riders were more active on weekends and summer afternoons, likely for leisure and sightseeing.

Trip Duration: Average ride duration for casual riders was ~39 minutes, compared to ~13 minutes for members.

Seasonality: Ride activity peaked in Q3 (summer) due to favorable weather and major city events.

Geographic Trends:

- The most active stations for casual riders were located near tourist hotspots like **Streeter Dr & Grand Ave** and **Millennium Park**.
- Members were concentrated around transportation and business districts such as **Canal St & Adams St**.

Strategic Takeaways

- **Target casual riders** with promotions around weekend passes, seasonal memberships, and loyalty discounts.
- **Highlight cost-saving benefits** and convenience of annual membership to frequent riders.
- **Deploy location-based campaigns** at high-traffic casual stations to convert tourists and leisure riders into members.

This analysis provides the Cyclistic marketing and operations teams with actionable insights to design data-driven campaigns that can **boost membership conversion rates** and **increase overall program profitability**.

Business Goal

The business goal is to understand how **annual members** and **casual riders** use Cyclistic bikes differently in order to identify opportunities to convert casual riders into annual members and increase profitability. By analyzing historical ride data and uncovering behavioral patterns, the Cyclistic marketing analytics team aims to provide actionable insights that can inform targeted marketing strategies and improve member conversion rates.

Preparing and Processing the Data for Analysis

Source of the Data and License Agreement

The data used in this project comes from Motivate International Inc., the operator behind Chicago's Divvy bike-share system, managed by Lyft Bikes and Scooters in partnership with the City of Chicago. It is publicly available under the Divvy Data License Agreement and used here solely for non-commercial, educational purposes. The dataset contains no personally identifiable information (PII).

Data Preparation and Standardization

The datasets include quarterly ride data from early 2019 through early 2020. Because the data format changed slightly in 2020, adjustments were made to align all files into a consistent structure. Only information directly relevant to the business objectives — such as trip duration, start and end times, and station details — was kept. Columns unrelated to the analysis were removed to simplify and standardize the data.

For the most recent quarter, some missing fields (like trip duration) were recalculated to match earlier data formats, ensuring accurate and consistent comparison across all periods. This preparation allows for smooth analysis of how Cyclistic's casual riders and annual members use the service differently.

Data Integrity

The datasets were obtained directly from the official “[Divvy-Tripdata](#)” source, ensuring their authenticity and reliability. To maintain consistency and accuracy, only data relevant to the business objectives was retained. Columns such as *bike ID*, *gender*, and *birth year* were excluded as they were not essential for this analysis. Records with missing or invalid values in key fields like *start time*, *end time*, *from station*, or *to station* were removed to ensure data completeness.

In addition, a small number of trips showing unrealistic durations (for example, lasting several days or weeks) were identified and removed, as these likely resulted from system or user errors. The *trip ID* column was standardized as a text identifier across all datasets to ensure compatibility between quarterly files and prevent technical inconsistencies. Overall, all tables were aligned to follow a uniform data structure, enabling accurate comparison and analysis across time periods.

Exploratory Data Analysis (EDA)

Overview

The goal of this analysis is to uncover behavioral differences between annual members and casual riders of Cyclistic. By exploring ride patterns across time, location, and usage metrics, we can identify actionable insights to help Cyclistic convert casual riders into annual members and improve overall profitability.

The cleaned and standardized dataset combines all five quarters of historical data (2019 Q1–2020 Q1), resulting in a total of 4,235,068 ride records for analysis.

User Composition

User Type	Total Rides	Percentage
Member	2,949,689	76.4%
Casual	911,302	23.6%

Annual members make up the majority of rides; however, casual riders still represent a significant segment of users. Their lower frequency but higher engagement makes them an important target group for conversion opportunities.

Ride Duration Behavior

User Type	Average Duration (minutes)
Member	13.97
Casual	39.96

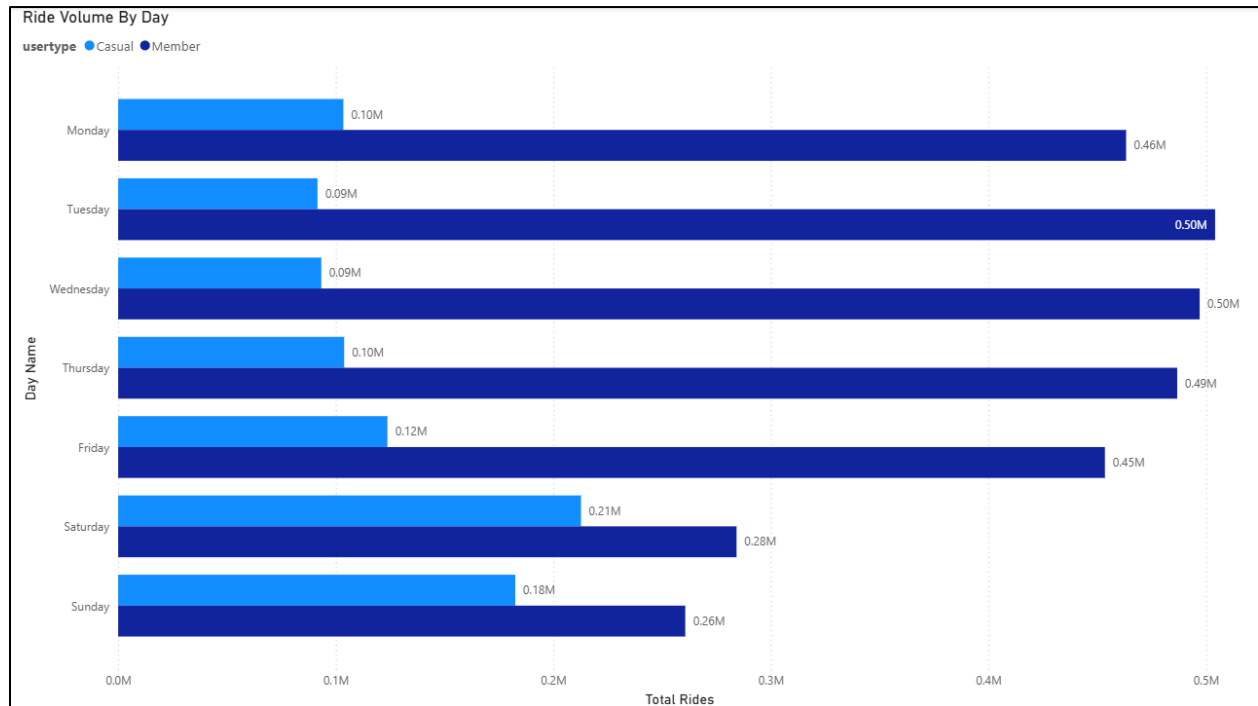
Casual riders take **trips roughly three times longer** than members. This pattern suggests that casual riders primarily use the service for **leisure, sightseeing, or social activities**, while members use it for **short, purpose-driven commutes** such as traveling to work or running errands.

Total Ride Time Contribution

User Type	Total Trip Duration (seconds)
Member	2,472,842,562
Casual	2,184,680,203

Despite being only 23% of total users, casual riders' **total ride duration nearly matches** that of members. This reinforces the finding that casual users, though fewer in number, spend considerably more time per ride — a behavior indicative of high engagement potential.

Weekly Usage Patterns

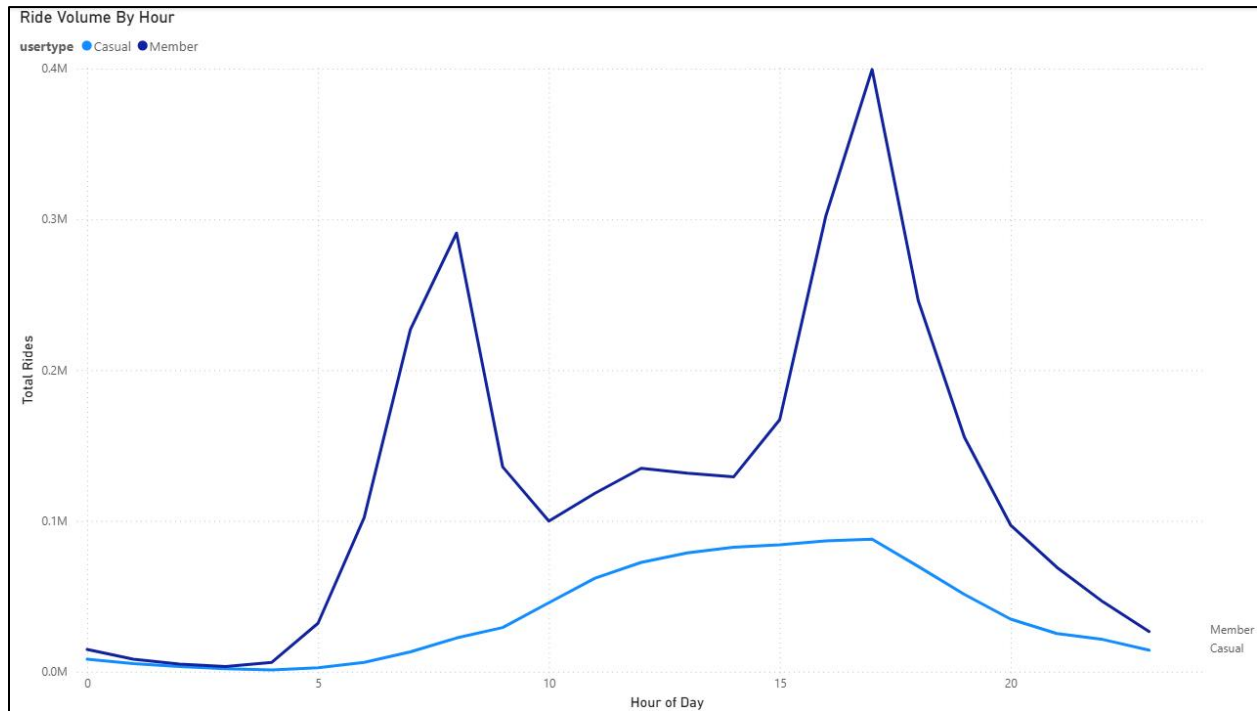


Casual riders are most active on **weekends**, while annual members show higher activity on **weekdays**.

This trend suggests that:

- **Members** rely on Cyclistic for **commuting** or **weekday routines**.
- **Casual riders** treat the service as a **recreational activity**, favoring weekends for exploration or leisure.

Hourly Usage Patterns



Analysis of start times reveals distinct usage trends throughout the day.

Annual Members:

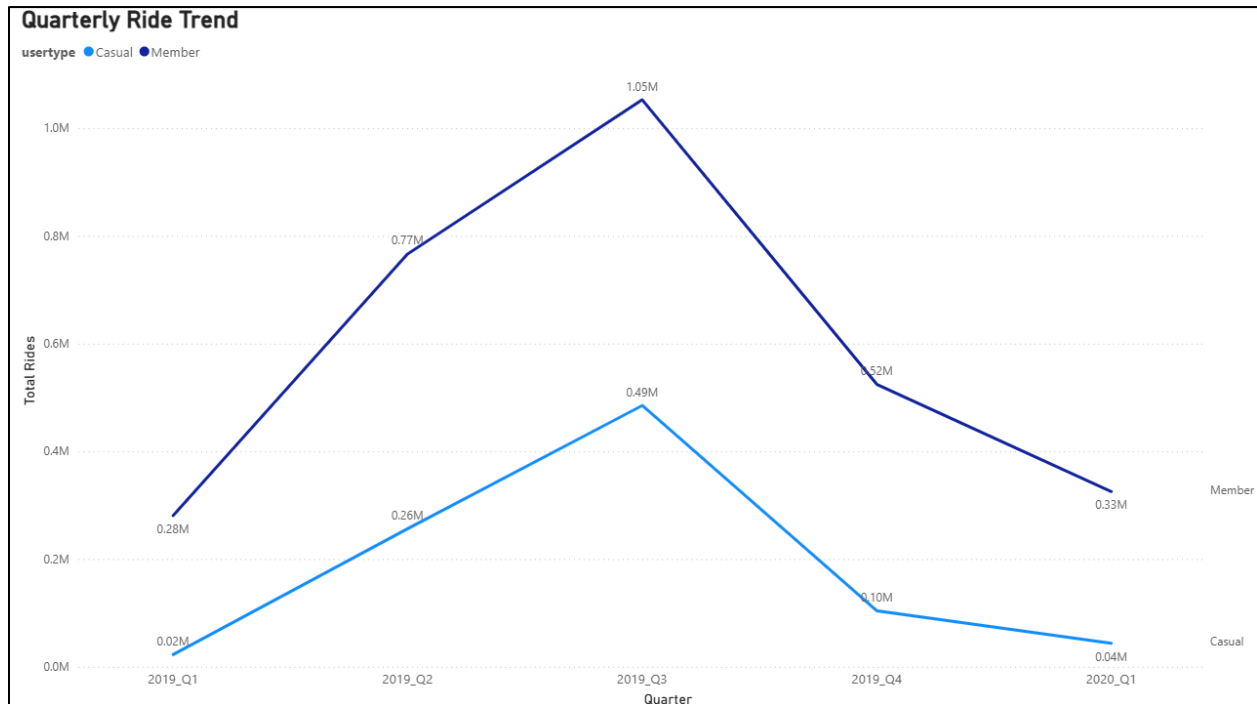
- Morning peak between **7–9 AM** and evening peak between **5–6 PM**.
- These times align with typical commute hours, suggesting that members primarily use Cyclistic bikes for daily transportation.

Casual Riders:

- Gradual increase starting from **late morning (around 10 AM)** and peaking between **3–5 PM**.
- This pattern reflects more **recreational or sightseeing** usage, especially during midday and weekends.

Marketing strategies could emphasize **weekday commuting benefits** for annual members, while promoting **weekend leisure rides** to attract casual users.

Seasonal and Quarterly Trends



Both user types show **sharp increases in Q3 (summer months)**.

This seasonal surge can be attributed to:

- **Favorable weather** encouraging outdoor travel.
- **City festivals and events** (e.g., Lollapalooza, Taste of Chicago) attracting tourists and local activity.
- **Increased leisure mobility** and lighter traffic, making bike-sharing more appealing.

Cyclistic can leverage this seasonal opportunity by promoting **short-term summer memberships** or **event-based discounts** to attract casual riders.

Popular Stations

Top Stations by User Type (Q3 2019)

Casual Riders	Total Trips	Member Riders	Total Trips
Streeter Dr & Grand Ave	31,618	Canal St & Adams St	17,127
Lake Shore Dr & Monroe St	20,602	Clinton St & Madison St	15,576
Michigan Ave & Oak St	13,122	Clinton St & Washington Blvd	14,663
Lake Shore Dr & North Blvd	12,553	Daley Center Plaza	12,366
Millennium Park	12,549	Franklin St & Monroe St	11,238

Observations:

- Casual riders primarily start trips near **tourist attractions** (Lakefront, Riverwalk, Millennium Park).
- Members favor **downtown business hubs** near Union Station and central offices.
- Average ride duration is longer for casual riders (43–49 minutes) at these popular leisure stations, compared to 11–14 minutes for members.

Cyclistic could deploy **targeted marketing at tourist-heavy stations**, offering incentives or QR-based discounts for casual riders to join membership programs.

Summary of Findings

Theme	Insight	Business Implication
Usage volume	Members dominate overall ridership	Maintain and enhance member retention strategies
Ride duration	Casual users ride 3× longer	High engagement potential; conversion opportunity
Day of week	Members = weekdays; Casual = weekends	Segment marketing by ride purpose
Hourly pattern	Members = commute peaks; Casual = midday leisure	Tailor promotions to time-of-day patterns
Seasonality	Peak in Q3 (summer)	Focus seasonal marketing on festivals and tourists
Location	Members = business zones; Casual = waterfront/tourist spots	Geo-target casual riders for membership offers

Key Takeaways

1. **Casual riders exhibit strong seasonal and recreational usage**, suggesting opportunities for **targeted membership promotions** during summer and weekends.
2. **Annual members demonstrate steady weekday commuting patterns**, showing satisfaction and consistent use of the service.
3. **Station-level insights** reveal a clear distinction between business and leisure travel zones, guiding **localized marketing strategies**.
4. By aligning marketing and membership campaigns with **time, location, and user behavior**, Cyclistic can effectively **increase conversion and profitability**.

Recommendations

Targeted Campaign Timing: Focus marketing campaigns during the **summer months (Q3)**, when ridership is highest, to encourage casual users to purchase memberships.

Commuter-Focused Messaging: Emphasize **convenience, cost savings, and flexibility** to retain and attract more annual members.

Leisure-Focused Promotions: For casual riders, highlight **fun, exploration, and family experiences**, particularly near tourist and waterfront stations.

Membership Incentives: Offer short-term membership trials or discounts during high-demand periods to convert occasional users into committed members.

Operational Planning: Allocate more bikes and maintenance support in high-demand stations during summer to ensure availability and customer satisfaction.

Appendix

Detailed SQL scripts, data-cleaning rules, and DAX measures are documented in the project's [GitHub repository](#) for transparency and reproducibility.